



PRIVATE AI RESEARCH SCOUT

Research Atlas

The complete user guide — from install to your first AI-ranked daily digest.
Everything runs locally on your own machine.

• Install & set up models

• Tune your research profile

• Scan, read & chat

A step-by-step guide

github.com/chemprojects/Research-Atlas

• OVERVIEW

What is Research Atlas?

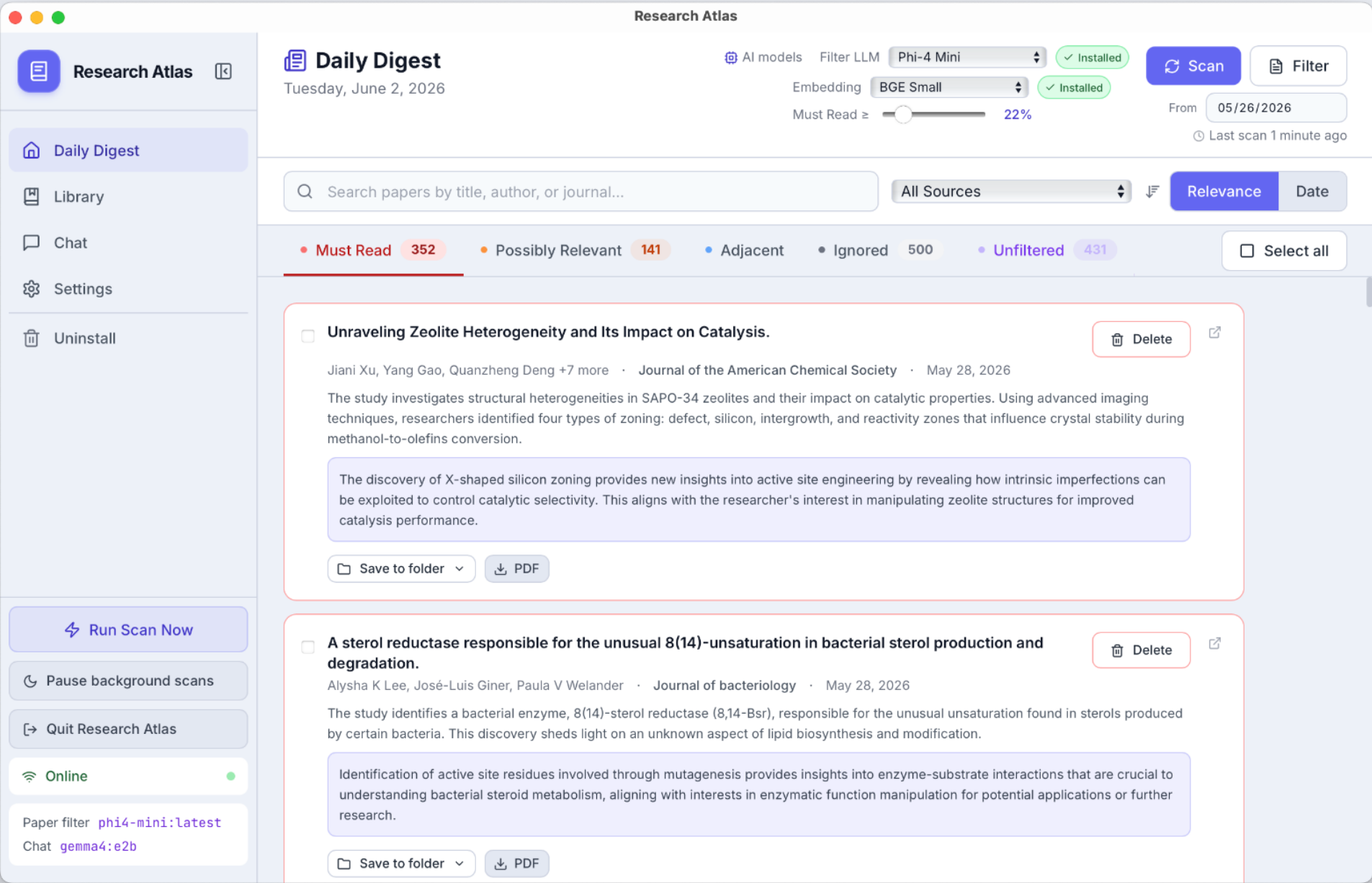
A desktop app that quietly watches the scientific literature for you, then ranks each new paper against **your** interests — so you open the app to a short, relevant reading list instead of an endless feed.

- 1

Scans scientific sources for you
Pulls fresh papers from OpenAlex, arXiv, PubMed, Crossref, Rxiv preprints and any peer-reviewed article publishers.
- 2

Ranks by relevance with local AI
On-device models score every paper into Must Read → Ignored tiers using your profile.
- 3

Reads, chats & cites — privately
Build a library, chat with papers, export citations. Nothing leaves your computer.



• THE WORKSPACE

Find your way around

Five places live in the left sidebar. The lower panel holds scan controls and shows which models are active at a glance.

- 

Daily Digest

Freshly scanned papers, sorted into relevance tiers.
- 

Library

Papers you saved, organised into folders.
- 

Chat

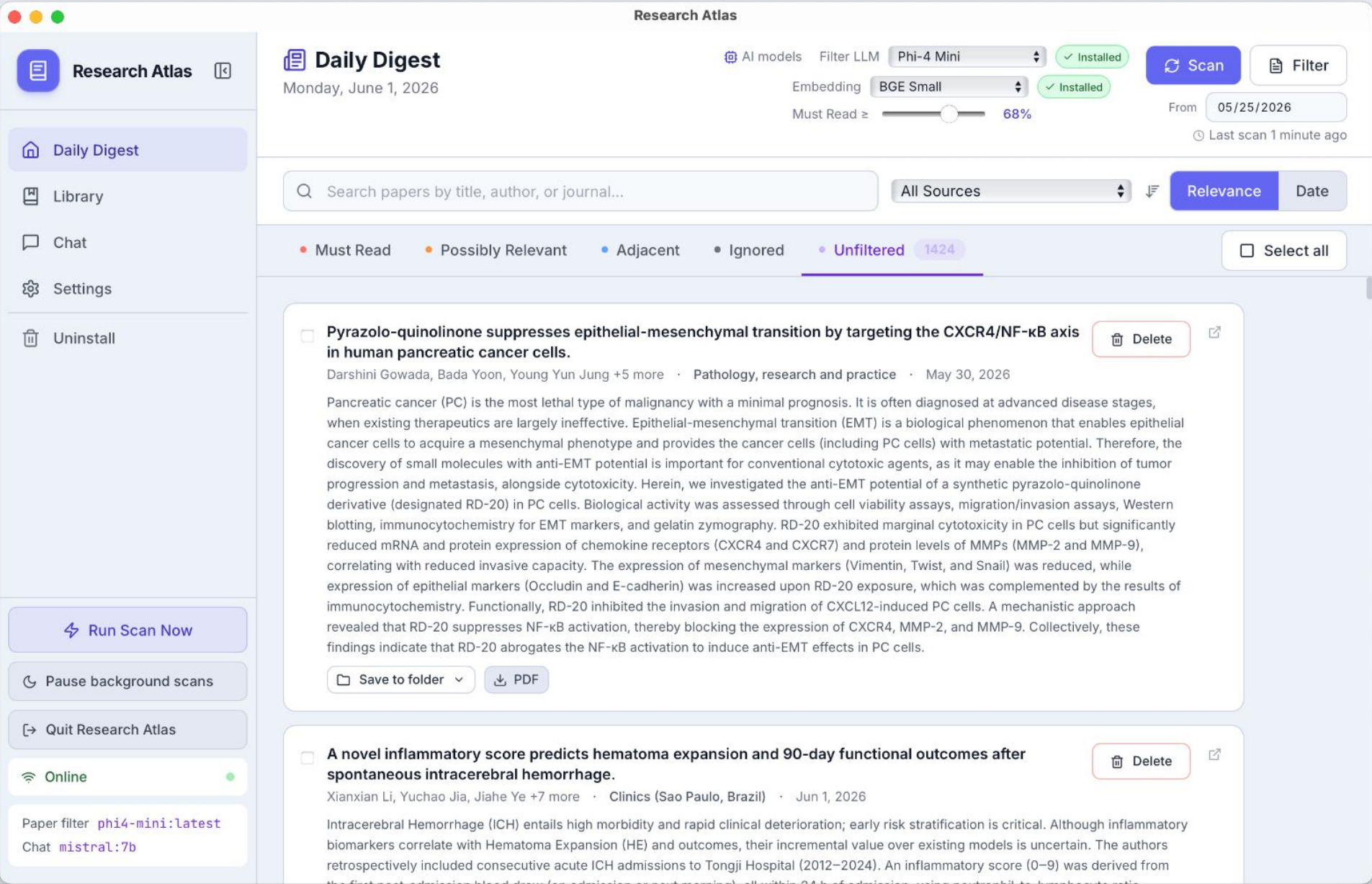
Ask questions across a folder, a project, or freely.
- 

Settings

Sources, research profile, AI models, hardware.
- 

Run Scan / Pause / Quit

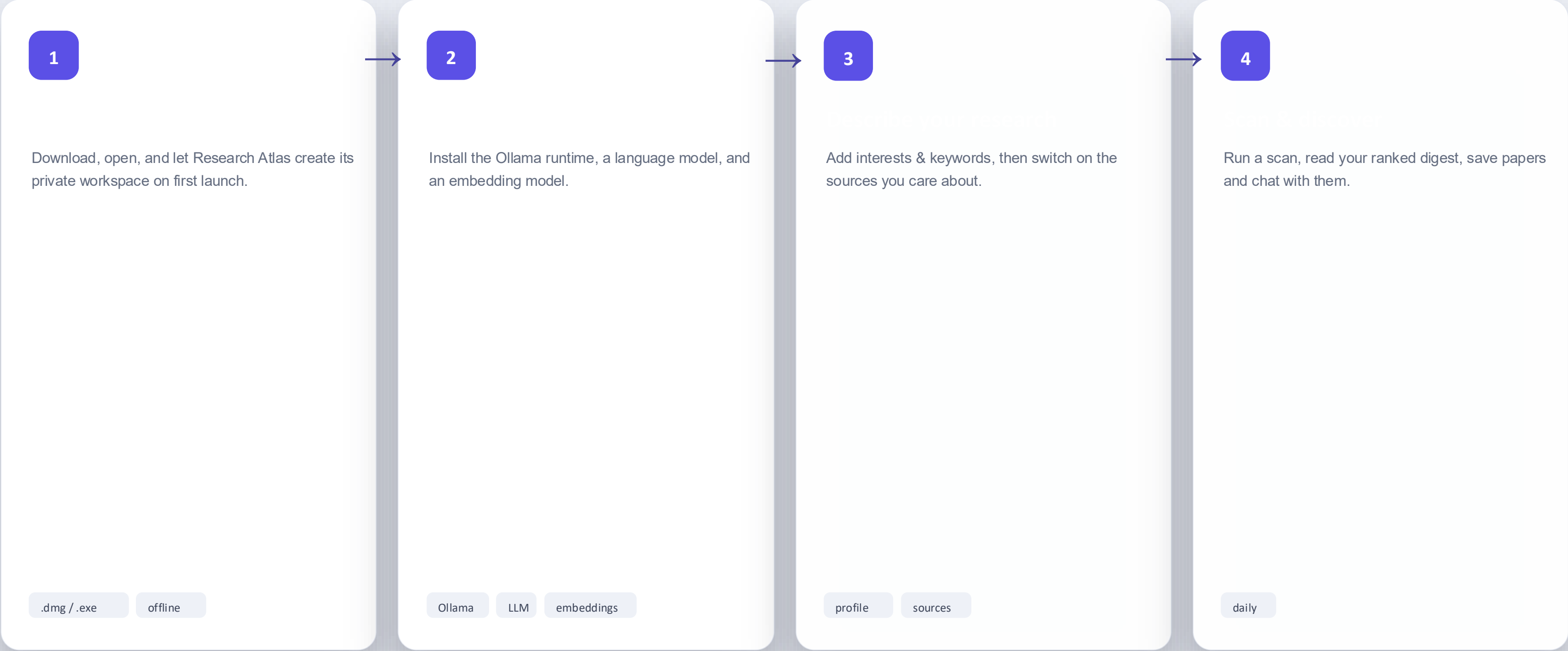
Bottom panel — plus an Online dot and the active filter & chat models.



• BEFORE YOU START

Your setup journey

Four short phases take you from a fresh install to a personalised digest. You only do the first three once.



1

PART ONE

Install & set up the AI engine

Get the app running, then install the local models that read and rank papers for you.

1 Install 2 Configure 3 Daily use

• STEP • INSTALL

Install & first launch

Grab the build for your platform from the project's releases, then open it like any desktop app.

- 1

Download the installer

macOS `.dmg` or Windows `.exe` from the project's Releases page.
- 2

Open it past the first-run prompt

macOS: right-click → Open

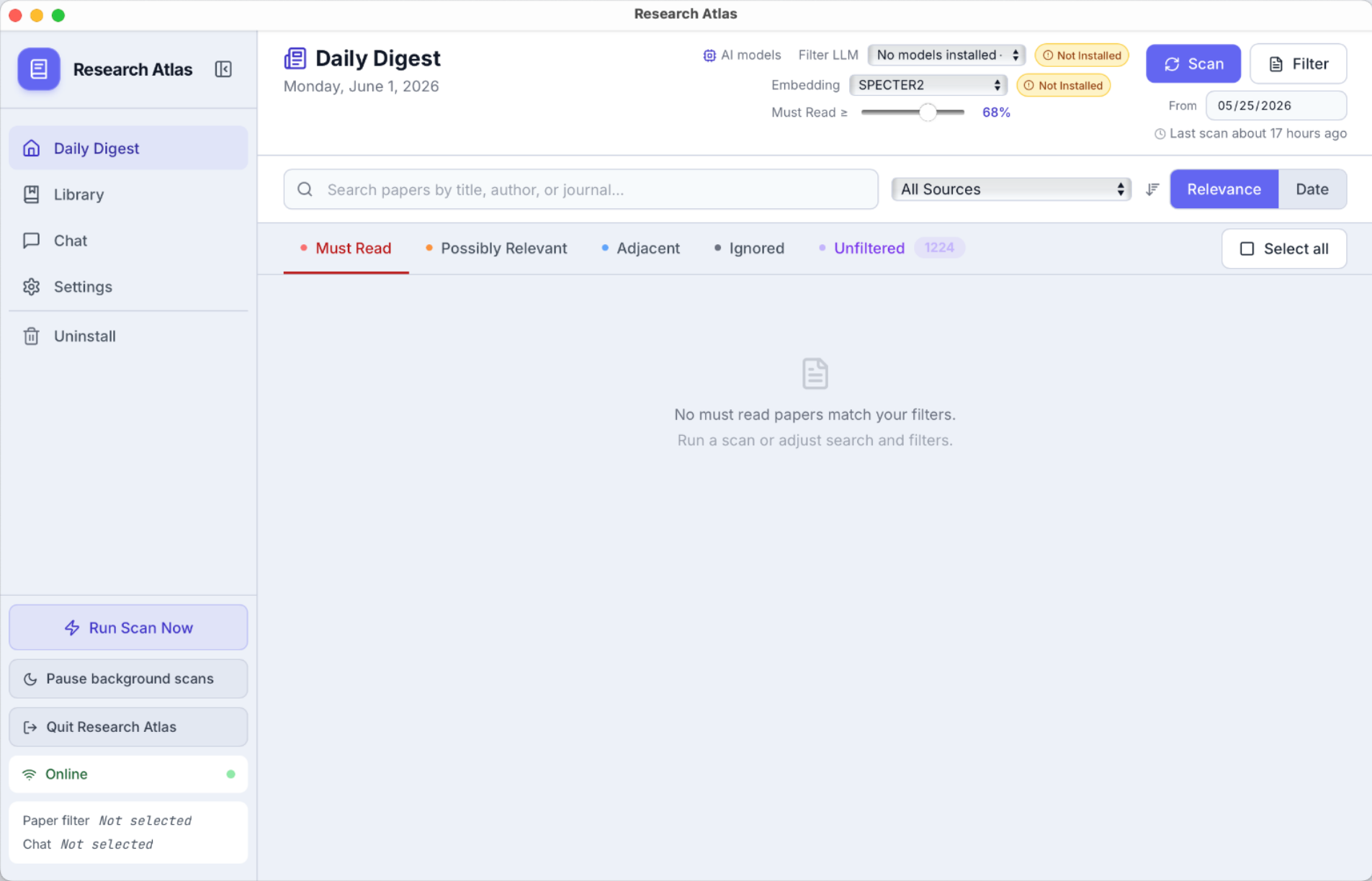
Windows: More info → Run anyway

on the SmartScreen notice.
- 3

Let it build the workspace

First launch creates a private folder & local environment — give it a moment.

! On a brand-new install the digest is empty and the top bar reads **“No models installed.”** That's expected — you'll fix it in the next three steps.



The engine: three pieces

Research Atlas runs AI locally through **Ollama**. You install it once, then add the models that do the thinking. Set them up in order.

1 Ollama runtime

The local engine that runs every model on your machine.
Must be installed and **Running** before anything else works.

2 Language model

Reads abstracts to filter papers and powers Chat.
Phi-4 Mini is light; **Mistral 7B** is a great all-rounder.

3 Embedding model

Turns papers into vectors for similarity & library search.
BGE Small is fast; **SPECTER2** is tuned for science.

★ **Hardware note.** Each model card lists its size and RAM/VRAM needs and flags what's **Recommended for Paper Filter** vs **Chat**. [Pick the largest your machine comfortably handles.](#)

STEP 1 OLLAMA RUNTIME

Install Ollama

Open **Settings** ► **AI Models** . The Ollama card sits at the very top — this is your foundation.

1 Click Install on the Ollama card

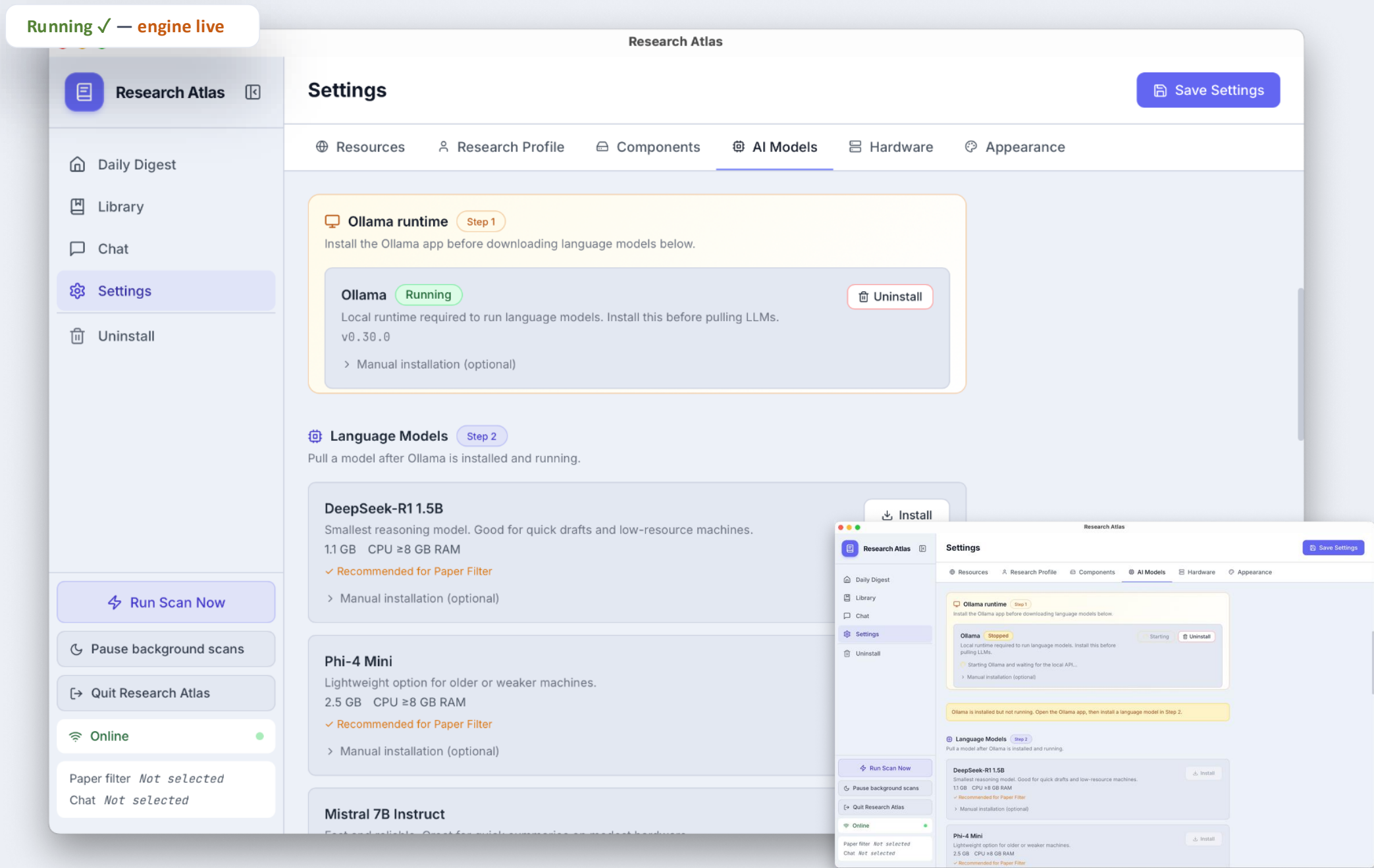
Or use
Manual installation
to point at an existing copy.

2 Wait for it to start

If you see **Stopped**
with a yellow notice, open the Ollama app so the local API comes up.

3 Confirm the green “Running” badge

A version like **v0.30.0** means the engine is live and Step 2 unlocks.



STEP 2 LANGUAGE MODEL MODEL

Add a language model

With Ollama running, pull at least one model and tell Atlas how to use it.

- 1

Install a model

Try [Phi-4 Mini](#) for filtering and [Mistral 7B](#) for chat. Keep the app open while it downloads.
- 2

Assign its job

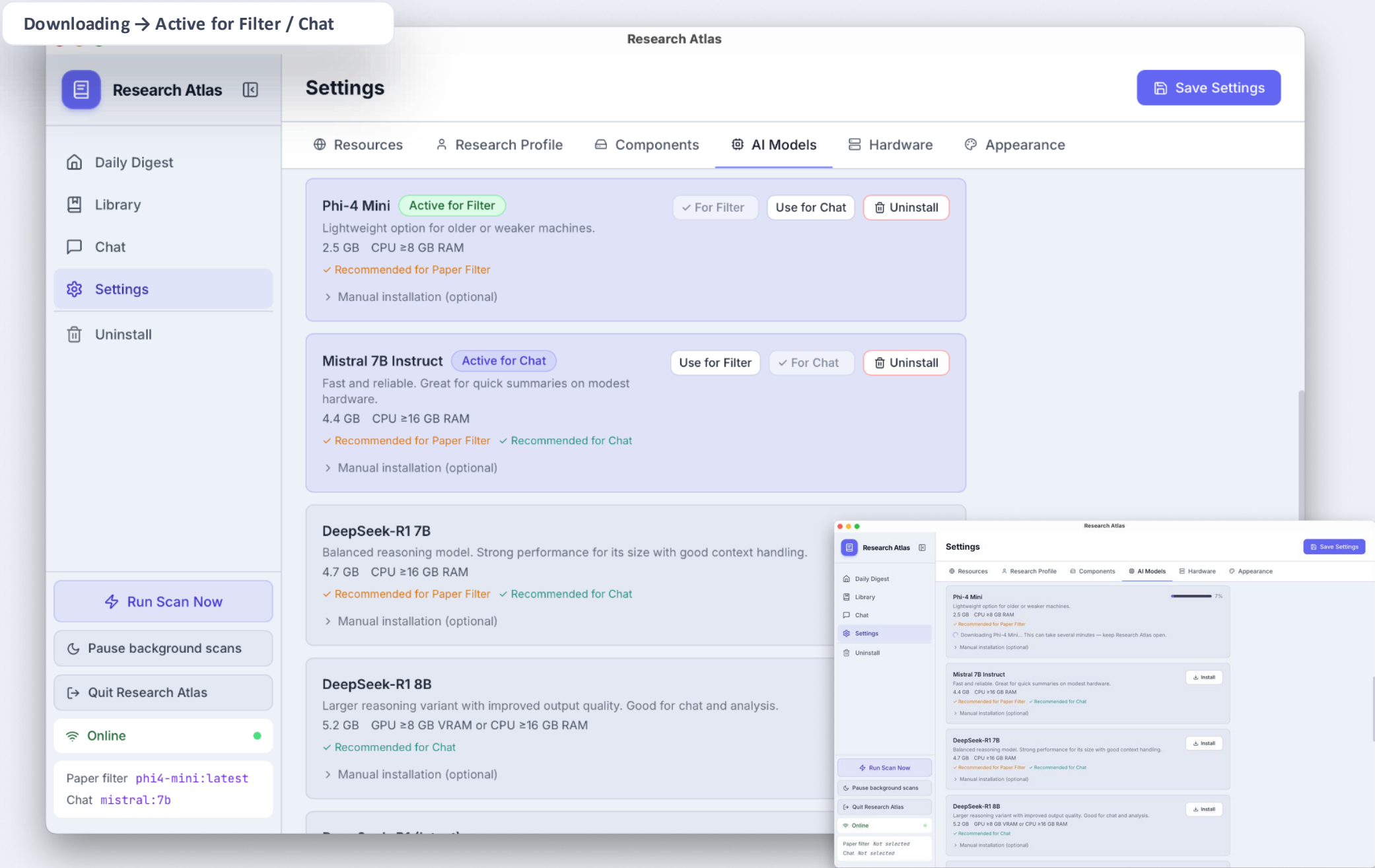
On an installed model click **Use for Filter** and/or **Use for Chat**

The active one shows a coloured badge.
- 3

Check the sidebar

The bottom panel confirms your picks, e.g. [Paper filter: phi4-mini](#)

[Chat: mistral:7b](#)



STEP 3 EMBEDDING MODEL MODEL

Add an embedding model

Scroll to **Embedding Models**. These power paper similarity and library search — independent of your chat LLM.

- 1

Pick one to install

[BGE Small](#) (0.1 GB, very fast) is the easy default. [SPECTER2](#) is purpose-built for research papers.
- 2

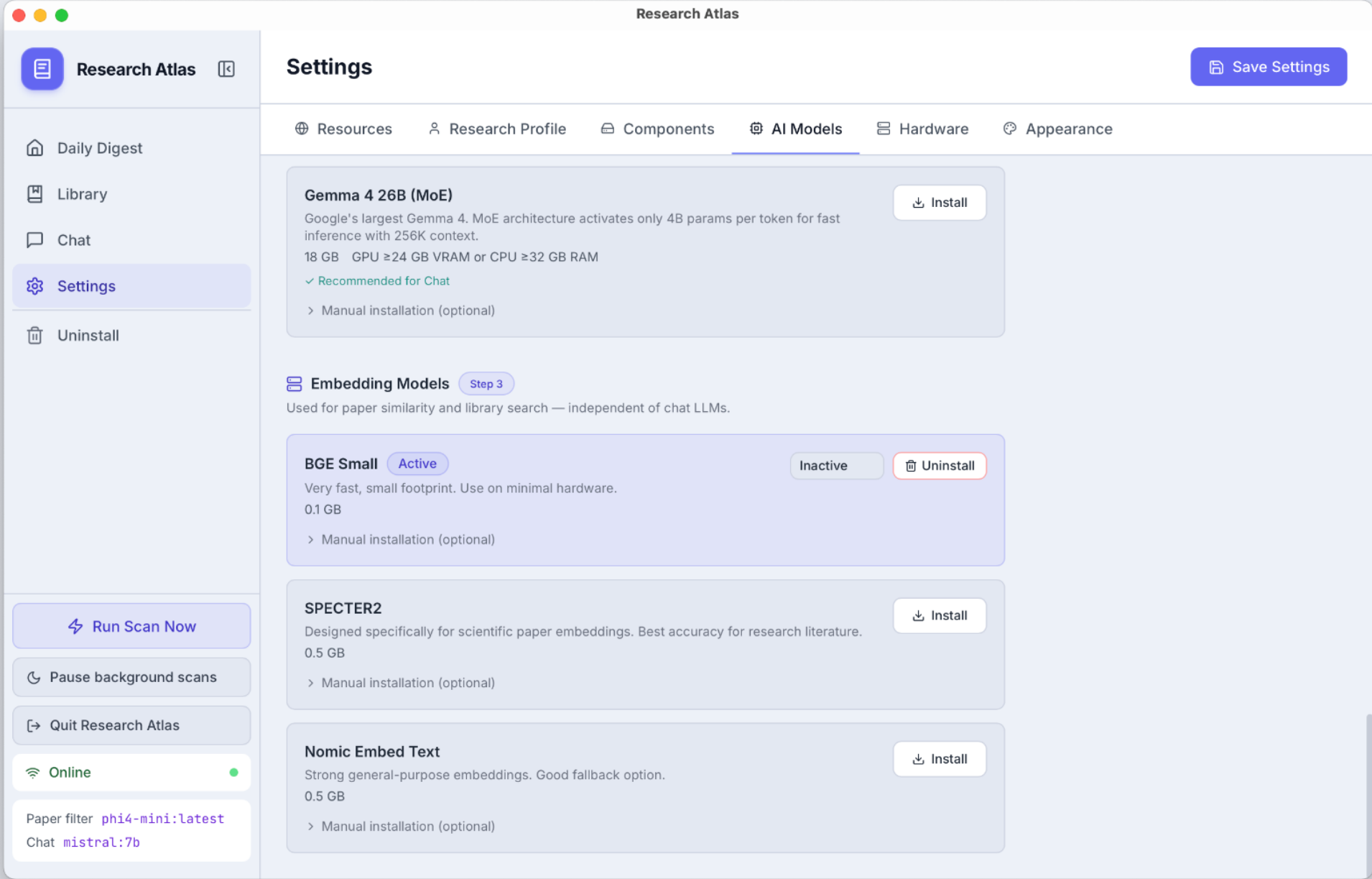
Make it Active

Once installed it shows an **Active** badge — only one embedding model runs at a time.
- 3

Save Settings

Use **Save Settings** (top-right) to lock in your full model setup.

✓ Engine complete: **Ollama running**, a language model assigned, and an embedding model active. You're ready to configure.



2

PART TWO

Tell Atlas what you study

Relevance ranking is only as good as your profile.
Spend five minutes here and every future scan gets sharper.

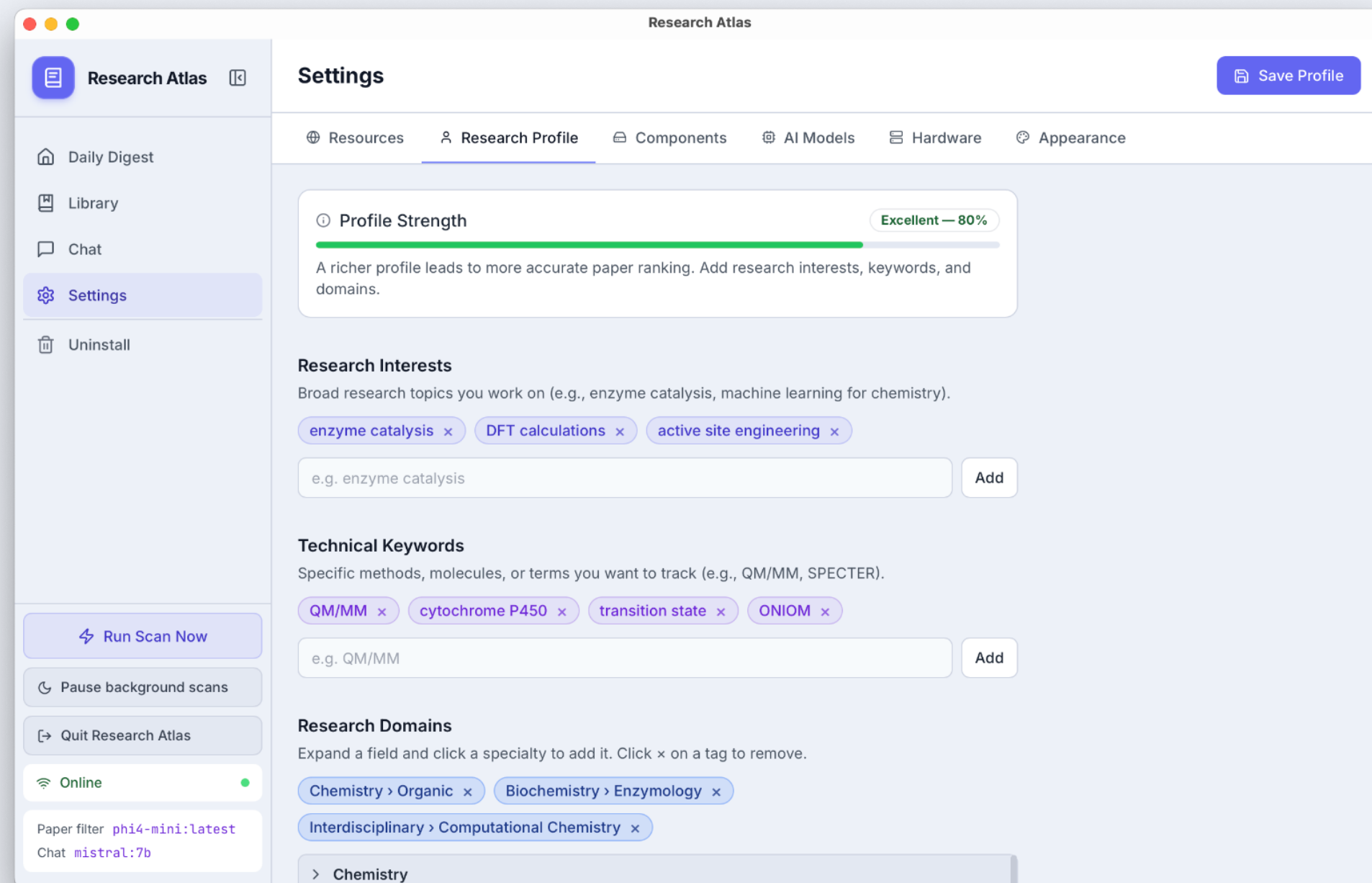
1 Install 2 **Configure** 3 Daily use

Build your profile

This is what the filter model compares every paper against. The richer it is, the better your ranking.

- 1 Add Research Interests**
Broad topics you work on — e.g. `enzyme catalysis`, `DFT calculations` .
- 2 Add Technical Keywords**
Specific methods or molecules to track — `QM/MM`, `cytochrome P450` .
- 3 Pick Research Domains**
Expand a field and click specialties to tag them. Watch **Profile Strength** climb.

★ Aim for **Excellent** strength, then hit **Save Profile** . You can refine anytime as your focus shifts.



• SETTINGS ► RESOURCES

Connect your sources

Choose which databases Atlas scans. Add yours or toggle on the ones that match your field.

- 1

Enable the right feeds

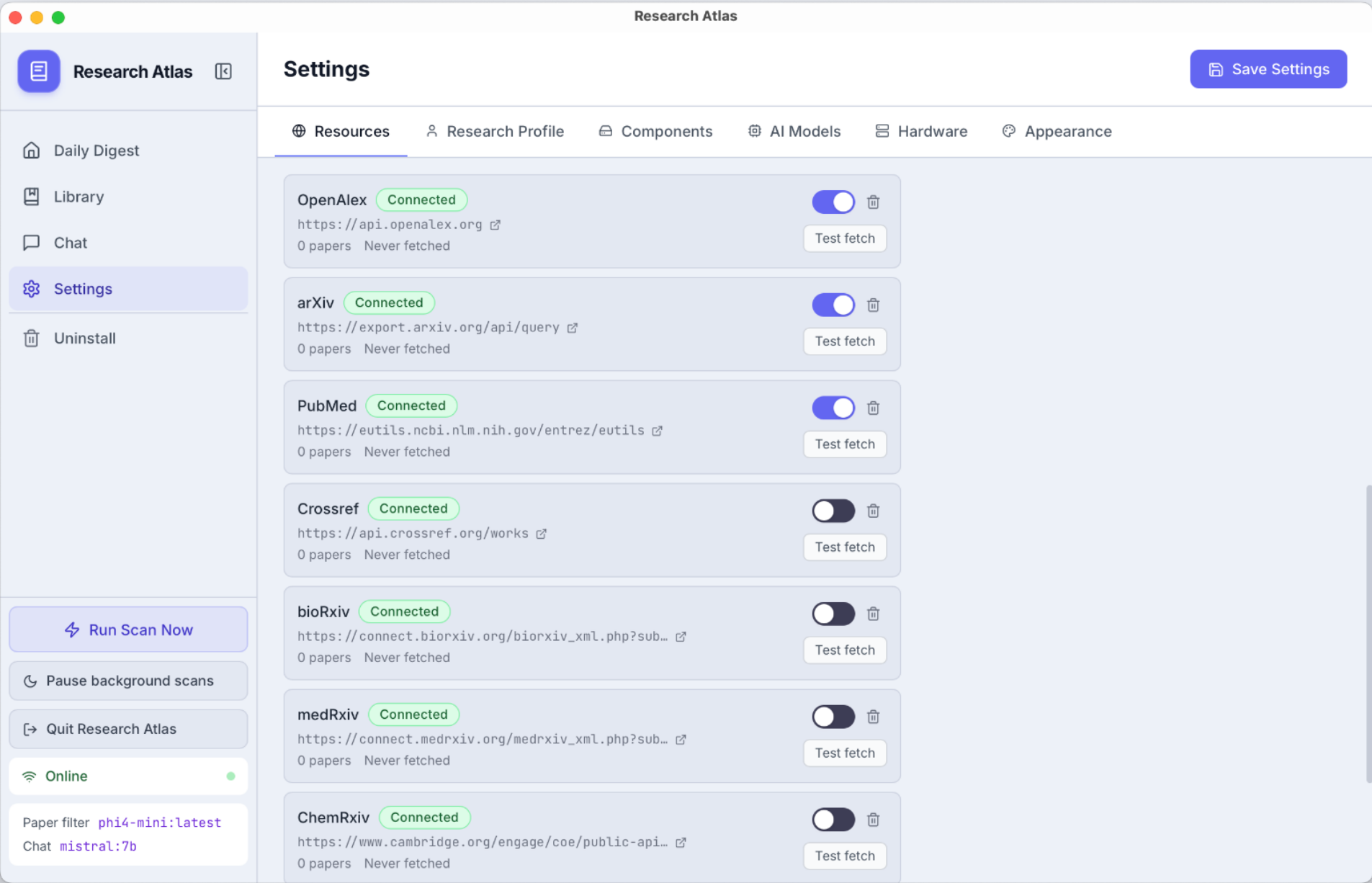
OpenAlex, arXiv and PubMed cover most ground; add bioRxiv, medRxiv, ChemRxiv or Crossref as needed. Or add yours.
- 2

Verify with Test fetch

Confirms the connection works before you rely on it in a scan.
- 3

Save Settings

Only enabled sources are queried, keeping scans fast and on-topic.



3

PART THREE

Your daily research workflow

Setup done. Now the part you'll repeat: scan, triage your ranked digest, save the keepers, and put AI to work.

1 Install 2 Configure 3 Daily use

• DAILY DIGEST

Run your first scan

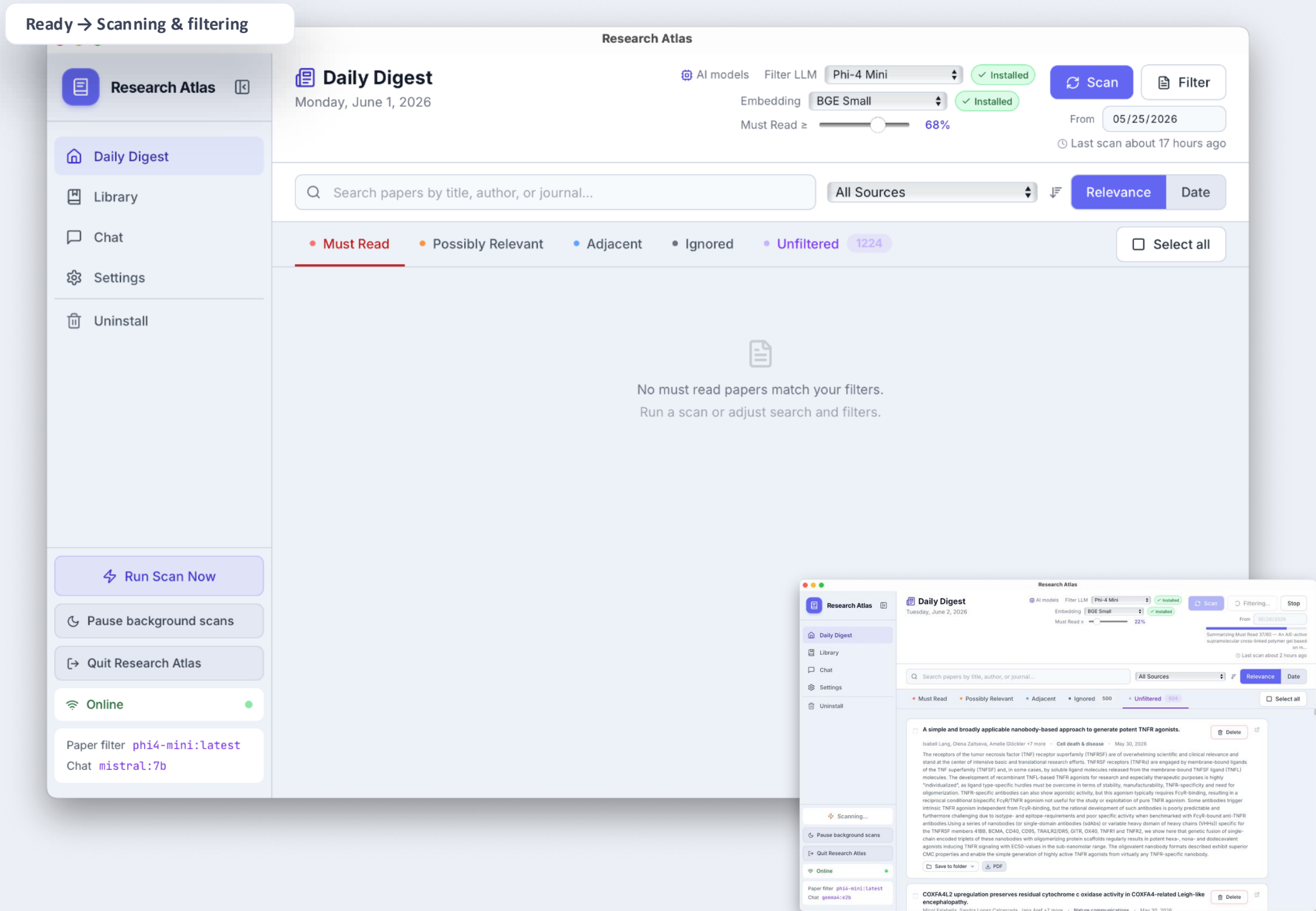
From the Daily Digest, set a date range and let Atlas pull and rank papers from your enabled sources.

1 Set “From” & press Scan
Or hit
Run Scan Now
in the sidebar. Confirm the badges read **Installed**.

2 Watch it work
Status moves through
Loading sources → Filtering → Summarizing

3 Tune the threshold
The
Must Read ≥
slider sets how strict the top tier is (e.g. 68%).

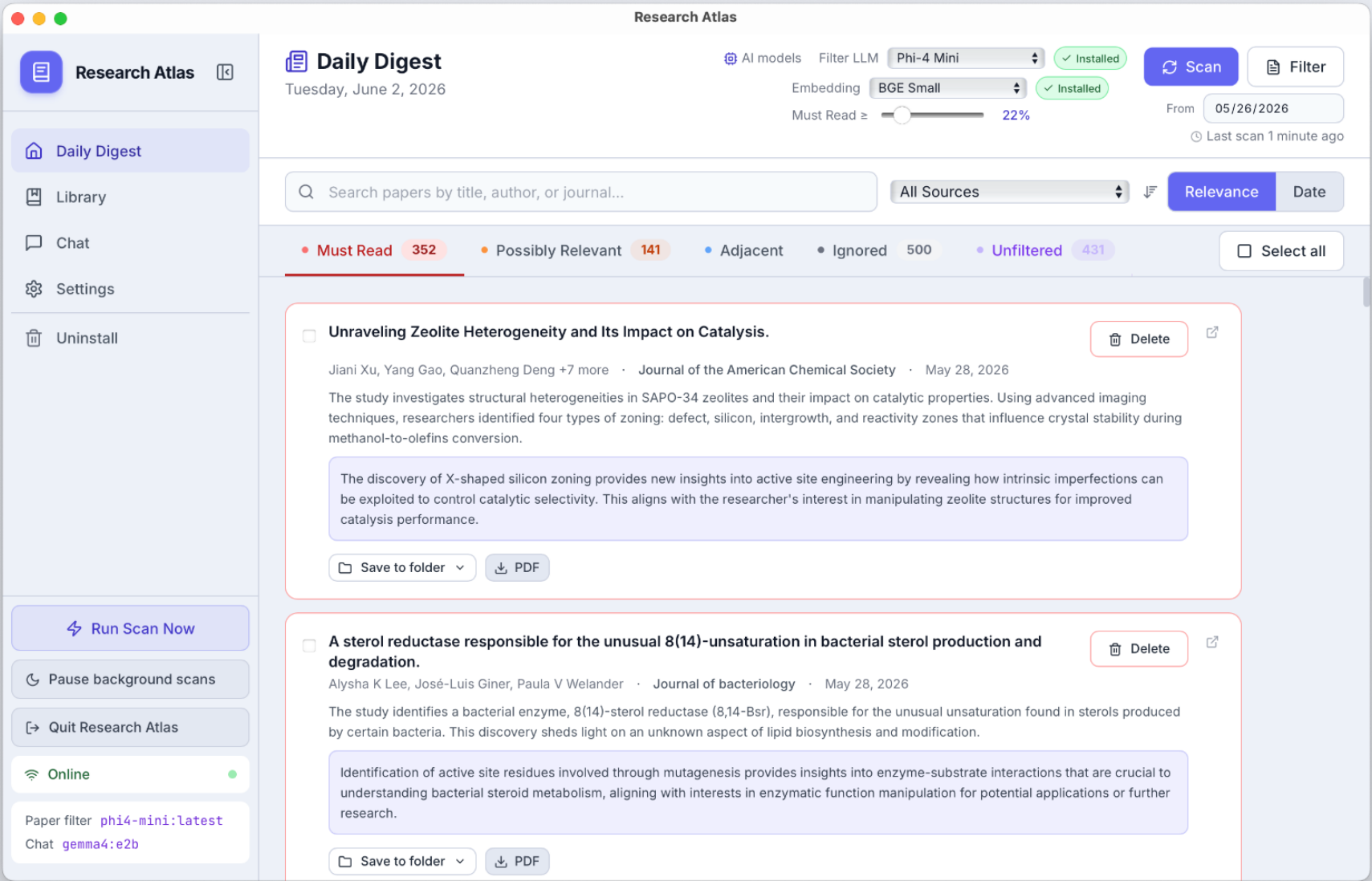
🔄 Leave background scans on and Atlas refreshes your digest on a schedule
— open it to a ready reading list.



• DAILY DIGEST

Read your ranked digest

Every paper lands in one of five tabs. Start at **Must Read** and work down only as far as you have time for.



● **Must Read**

Top matches above your threshold. Your priority queue.

● **Possibly Relevant**

Solid matches worth a scan when you have a moment.

● **Adjacent**

Nearby fields — good for staying broadly aware.

● **Ignored**

Set aside as off-topic. **Reset Ignored** to revisit.

● **Unfiltered**

Raw pulled results before AI scoring is applied.

• DAILY DIGEST

Browse & refine

Each result shows title, authors, journal, date and a short summary — with the matching reasons highlighted.

- 1

Search & filter

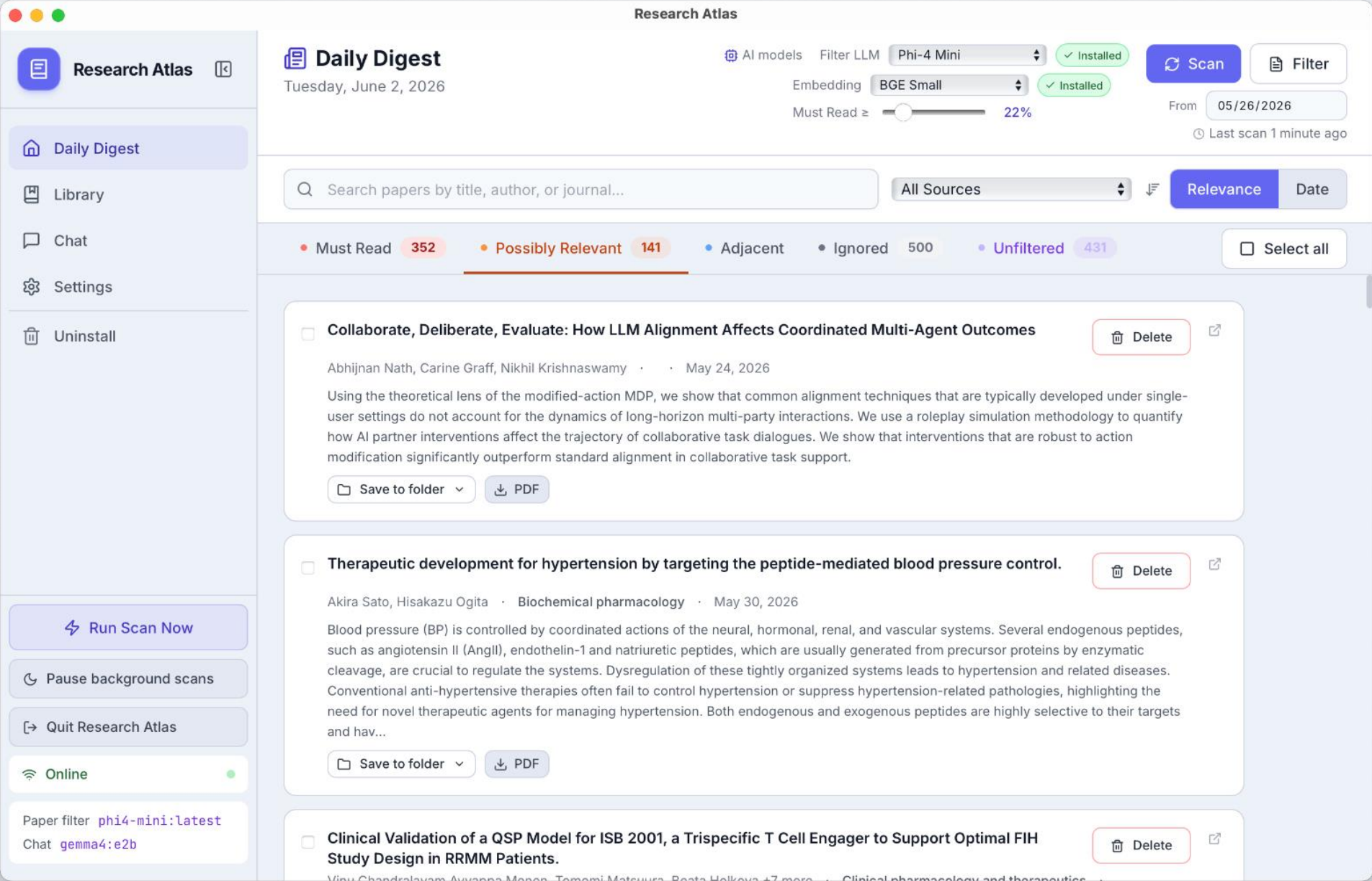
Search by title, author or journal; narrow by source with the **All Sources** dropdown.
- 2

Sort your way

Toggle **Relevance** or **Date** to reorder the list.
- 3

Act on a paper

Grab the **PDF**, open the source, **Delete**, or **Select all** to act in bulk.



• DAILY DIGEST → LIBRARY

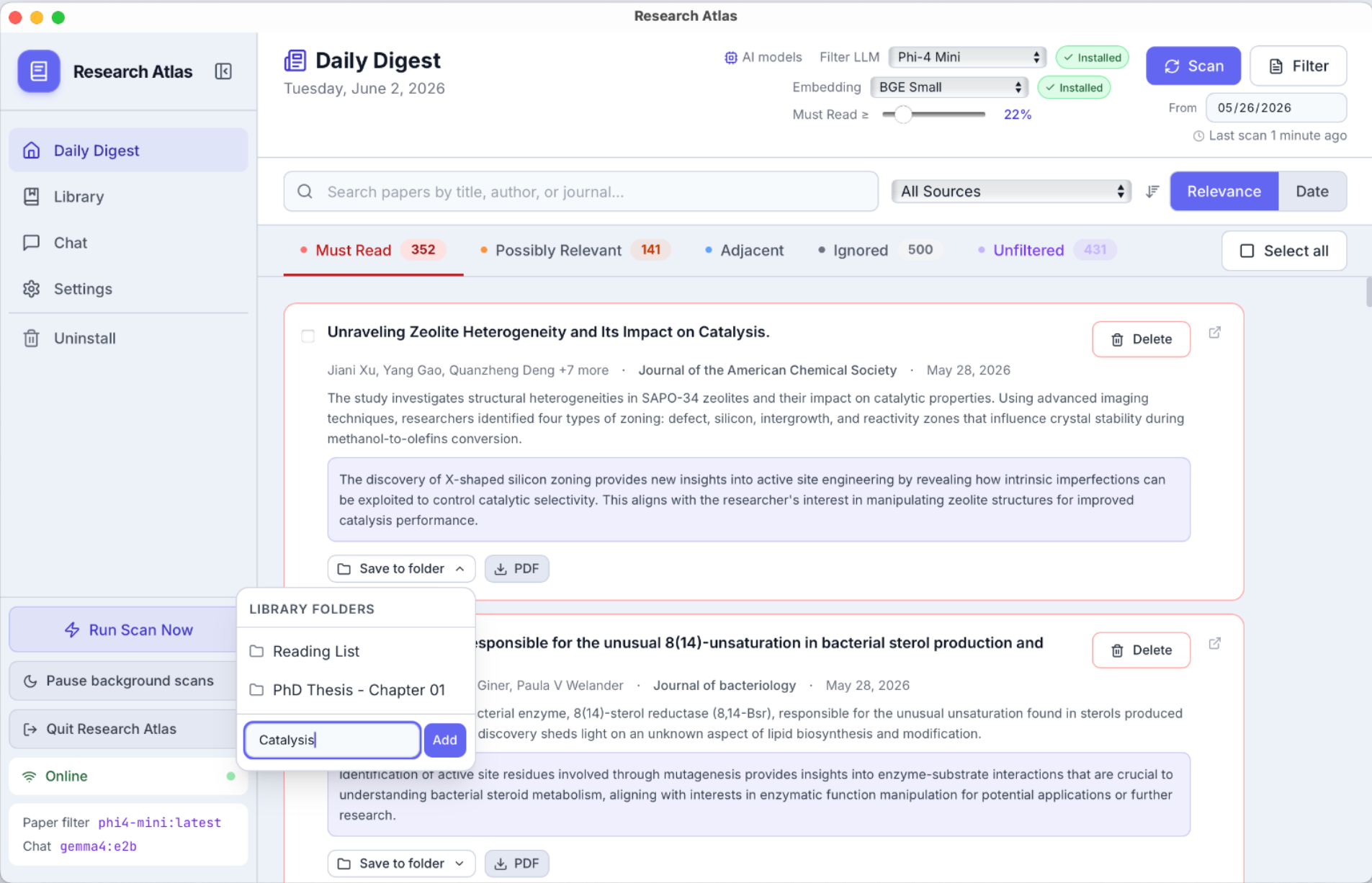
Save papers to folders

Found a keeper? File it into your Library so it's there when you need it.

1 **Open Save to folder**
On any result, click the **Save to folder** dropdown.

2 **Pick an existing folder**
e.g. `ReadingList` or `PhD Thesis — Chapter 01`.

3 **Or create one inline**
Type a name and hit **Add** — no need to leave the digest.



• LIBRARY

Your Library

Everything you saved, grouped by folder.

Click any paper to read its abstract and details on the right.

- 1

Switch folders & search

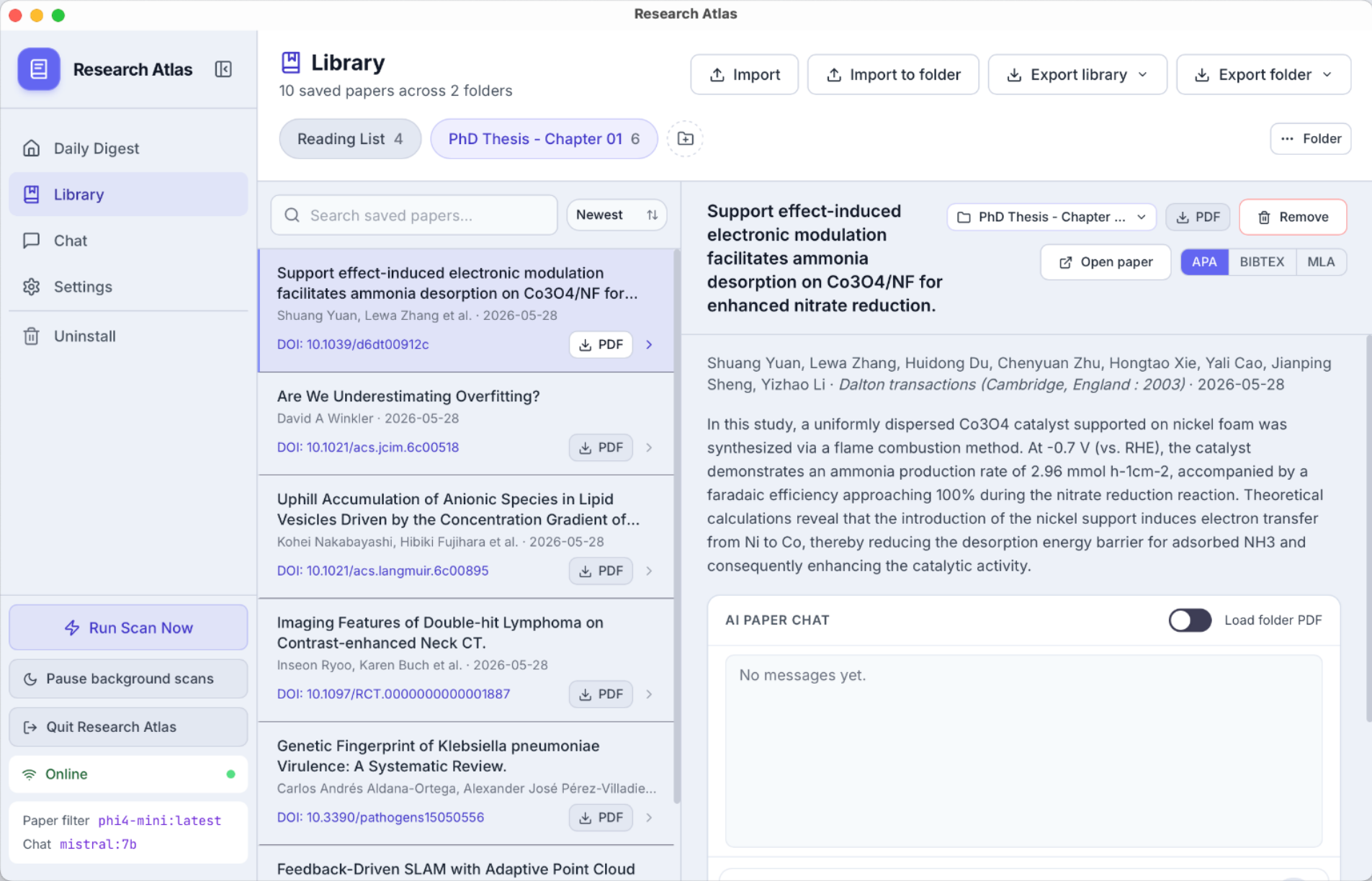
Use the folder chips up top; search saved papers and sort **Newest / Oldest**
- 2

Open the detail pane

Read the abstract, **Open paper**, grab the **PDF**, move it, or **Remove** it.
- 3

Import & export

Import your own papers, and **Export library** or **Export folder** to share.



- **LIBRARY ▶ AI PAPER CHAT**

Chat with a paper

Each saved paper has its own assistant — ask for a summary, the methods, or how it relates to your work.

- 1

Load the full PDF

Flip **Load folder PDF**

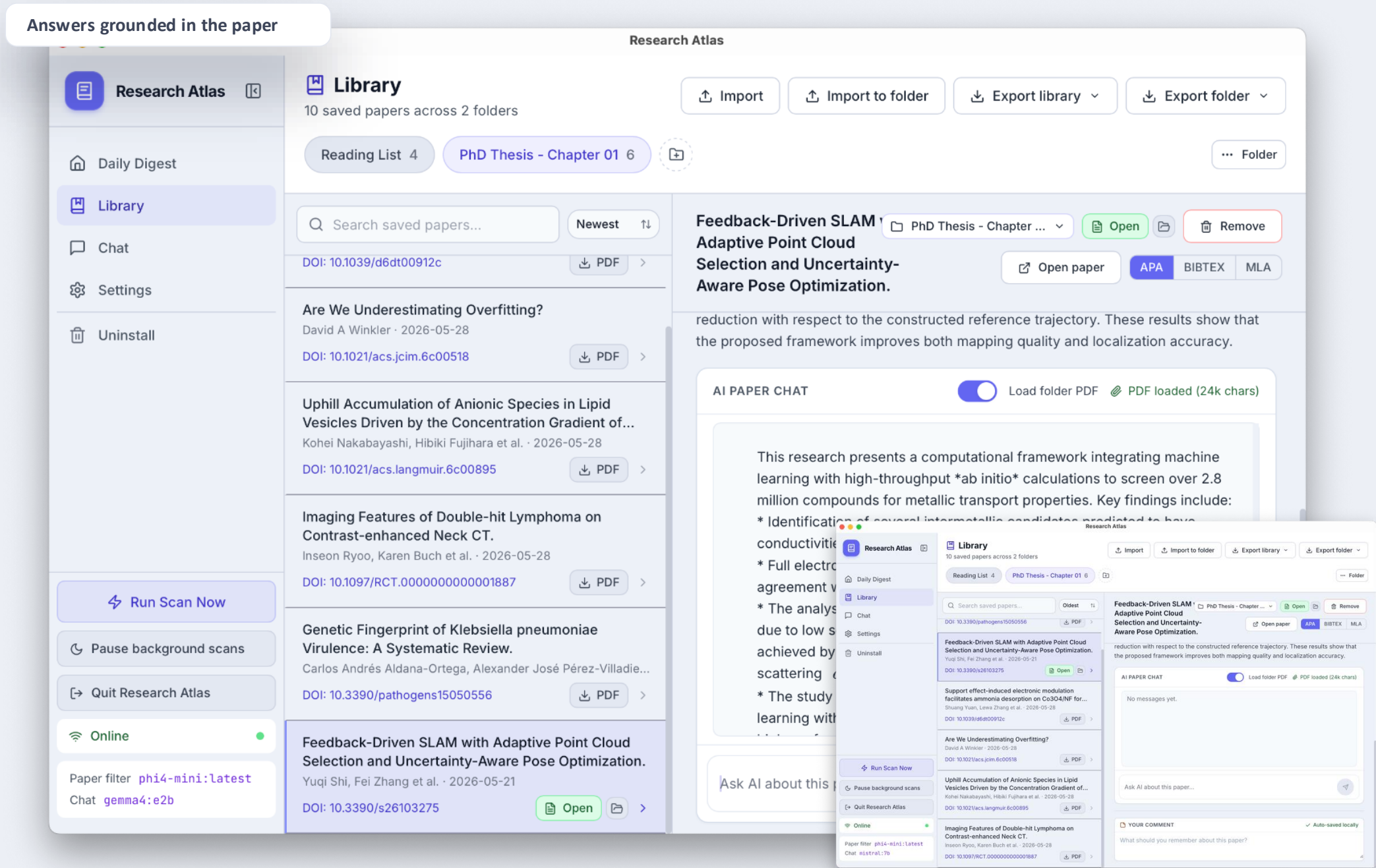
on so answers draw on the full text, not just the abstract.
- 2

Ask anything

Type into **Ask AI about this paper**, or tap a suggestion like **Summary of the paper**
- 3

Jot a private note

The **Your Comment** box auto-saves locally — perfect for “why this matters”.



• LIBRARY ▶ CITATIONS

Cite in one click

Need a reference for a manuscript or thesis? Atlas formats it instantly from the detail pane.

- 1

Choose a style

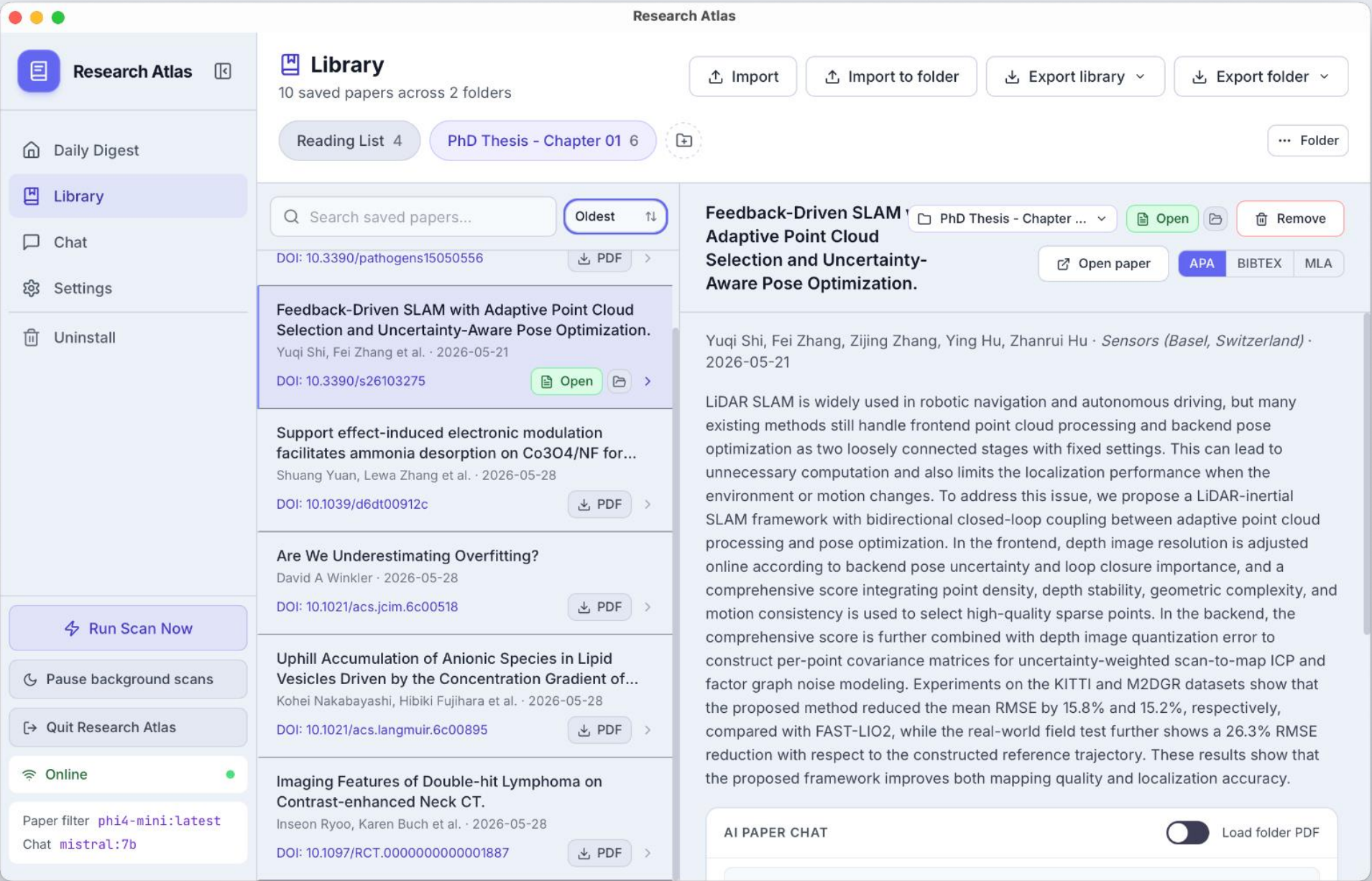
Switch between [APA](#), [BIBTEX](#) and [MLA](#) at the top of the paper.
- 2

Copy the citation

Drop it straight into your reference manager or document.
- 3

Export in bulk

Use **Export folder** or **Export library** to take a whole reading list with you.



• CHAT

Chat across your research

Beyond a single paper, the Chat tab reasons over whole folders — or just thinks alongside you.

- 1
- Pick a mode**

Library (chat a folder),

Projects (grouped work), or

Free Chat (general questions).

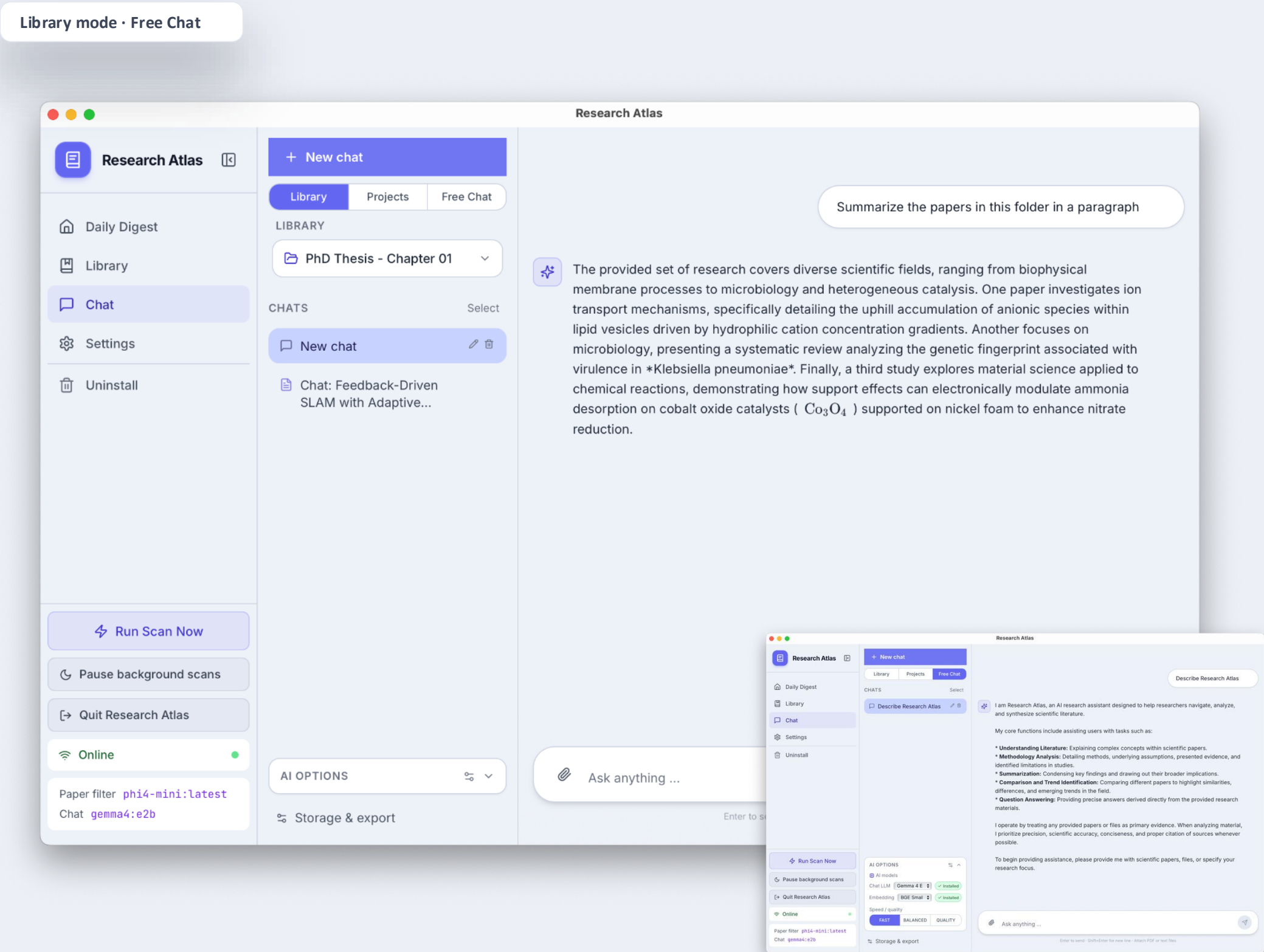
- 2
- Choose a folder**

In Library mode, select e.g. **PhD Thesis —Chapter 01**, then ask “summarize these papers”.

- 3
- Set AI Options**

Switch the **Chat LLM** & **Embedding**, and trade

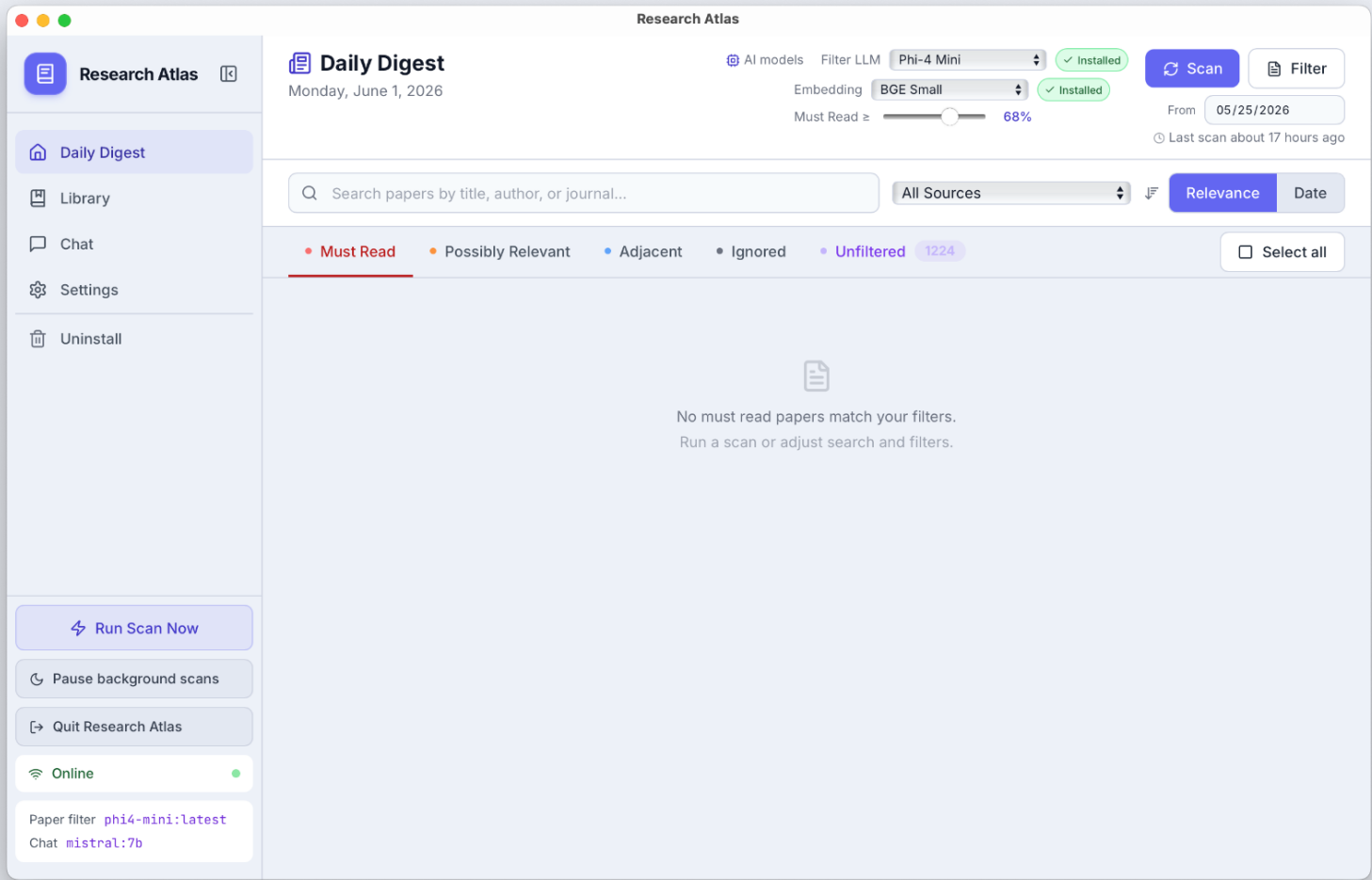
Fast ⇌ Balanced ⇌ Quality



• SIDEBAR REFERENCE

Everyday controls

The lower sidebar is your control panel — scanning, background behaviour, connection, and a live read on which models are doing what.



Run Scan Now

Trigger an on-demand scan whenever you want fresh results.



Pause / Resume

Stop background scans while you focus, resume later.



Online & models

Connection dot plus the active **Paper filter & Chat** models.



Quit / Uninstall

Close cleanly, or remove Atlas and its models from Settings.

• QUICK REFERENCE

Your setup, at a glance

Once these are ticked off, Research Atlas runs on its own — just open it and read.

- ✓

App installed & launched
Workspace created locally on first run.
- ✓

Models assigned
A language model for Filter/Chat + an active embedding model.
- ✓

First scan complete
Papers ranked into Must Read → Ignored.
- ✓

Ollama running
Green badge in Settings ▶ AI Models.
- ✓

Profile & sources set
Interests, keywords, domains — and the feeds you care about.
- ✓

Library started
Keepers saved into folders, ready to cite & chat.



You're all set.

Open Research Atlas each morning to a short, ranked reading list — built privately, on your machine, around exactly what you study.

Keep it fresh

Leave background scans on so the digest is ready when you are.

Refine often

Add keywords as your focus shifts — ranking sharpens over time.

Mind your hardware

Bigger models read better; match them to your RAM/VRAM.

github.com/chemprojects/Research-Atlas